

Sajag AI (सजग)

An on-device, multimodal scam shield for Snapdragon-powered HP AI PCs

Participant	Tanmay Sahare, B.Tech Aerospace Engineering, IIT Kanpur (individual)
Code	https://github.com/tanmaysahare/sajag-ai (Apache-2.0)
Target platform	Windows 11 on Snapdragon X / X2 Elite (HP OmniBook and other HP AI PCs), Hexagon NPU via ONNX Runtime QNN Execution Provider
AI models	Qualcomm AI Hub: Silero-VAD, Whisper-Small (and Whisper-Large-v3-Turbo), Llama-3.2-3B-Instruct (optional). Open source: PP-OCRv4 (PaddleOCR / RapidOCR). Own: ONNX scam-intent classifier
Languages	English, Hindi (Devanagari) and Hinglish, in speech, on screen and in alerts

1. Summary

Sajag AI watches the three places a modern scam actually happens on a PC: the call audio (any app, via WASAPI loopback), the screen (OCR of fake warrants, scareware pop-ups and payment pages) and the system (remote-access tools such as AnyDesk starting during a call). Every model runs locally on the Snapdragon Hexagon NPU. A multilingual risk engine and a temporal, cross-modal fusion model turn those signals into a verdict (SAFE, CAUTION, WARNING, DANGER). When the risk crosses WARNING, Sajag interrupts the user with a full-screen bilingual card, a spoken warning, the exact reasons ("asked you to stay on the call and not tell anyone") and the national cyber-fraud helpline 1930. Nothing is uploaded and nothing is stored.

Results at a glance: On a reproducible synthetic evaluation Sajag detects 91% of 200 scam calls with 0.5% false alarms on 200 benign calls (median 2 turns to the first warning), passes all 5 end-to-end scenarios, and 27 of 27 tests. In the digital-arrest scenario it raises a WARNING at 20 s and DANGER at 26 s, before the transfer request at 45 s.

2. Problem

Indians lost ₹22,845 crore to cyber fraud in 2024, a 206% increase over 2023 (MHA). Digital-arrest scams grew from 39,925 cases (₹91 crore) in 2022 to 1,23,672 cases (₹1,918 crore) in 2024 (I4C data reported by ThePrint). Awareness campaigns cut cases by 86% in 2025, but the average loss per remaining case rose from ₹1.55 lakh to ₹3.73 lakh. Microsoft found 7 in 10 Indian consumers met a tech-support scam in a year, the highest rate in the world, and McAfee found 47% of Indians had faced or knew someone who faced an AI voice-clone scam.

These scams increasingly run on the PC: victims are held on WhatsApp Desktop, Skype, Teams or Zoom video calls, shown forged CBI documents on screen, persuaded to install AnyDesk or TeamViewer (both the subject of RBI and NPCI warnings) and walked through net banking. Every current defence works before the call (awareness ads, bank SMS) or after it (the 1930 helpline, cybercrime.gov.in). Nothing protects the person during the call, because doing that in the cloud would require streaming their screen and conversations to a server.

3. Solution

Component	What it does	Runs on
Call sentinel	WASAPI loopback + microphone capture from any call app; Silero-VAD gate; Whisper-Small ASR with a forced < hi > / < en > decoder prompt so Hinglish is transcribed rather than translated	Hexagon NPU
Screen sentinel	Screen capture every 2 s; a 64-bit difference hash skips unchanged screens; PP-OCRv4 detection + recognition through static-shape NPU graphs	Hexagon NPU
System sentinel	Detects 14 remote-access tools, 9 call apps and payment screens (UPI PIN, add payee, OTP)	CPU (microseconds)
Risk analyzer	300+ cues across 10 tactics in English, Devanagari and Hinglish; awareness damping; ONNX scam-intent classifier (8 classes)	NPU / CPU

Fusion	Decaying tactic memory across modalities; escalation rules for remote access + OTP, secrecy + authority + money, pay-to-receive; 7 scam families	CPU
Explain + alert	Bilingual reasons and family-specific advice; optional Llama-3.2-3B via Genie; full-screen interrupt, voice, toast, local dashboard	NPU / CPU

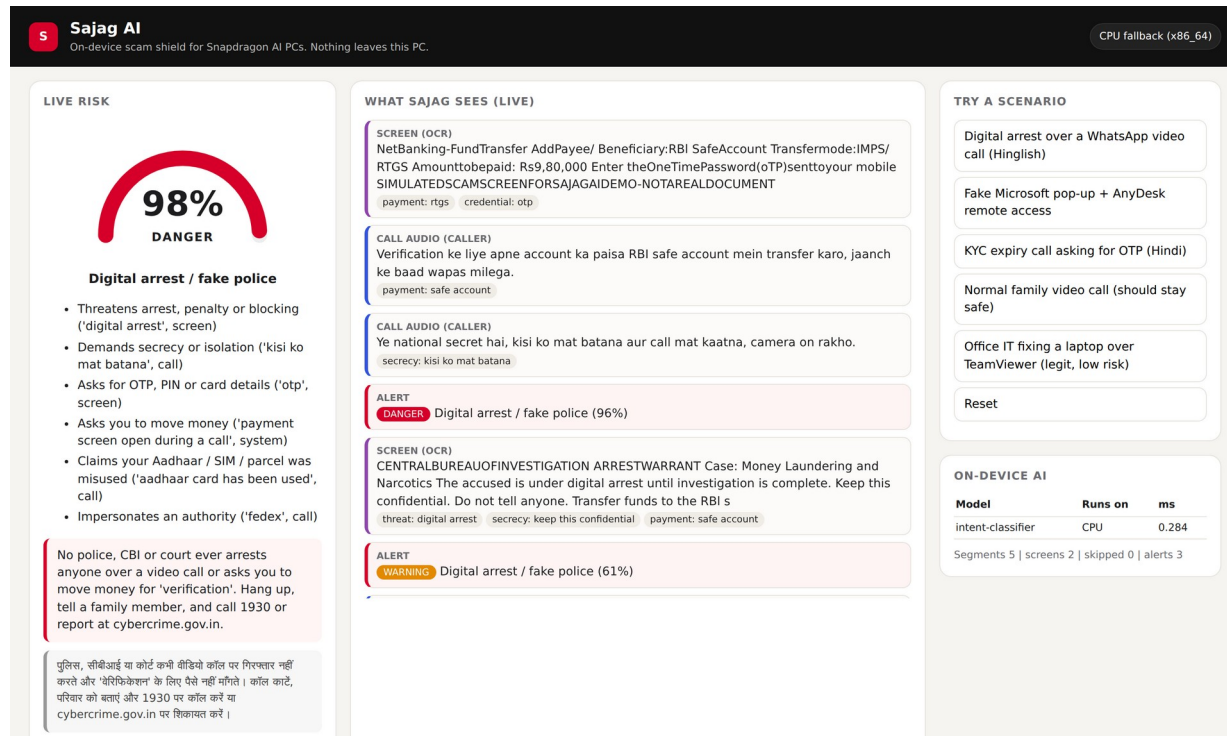


Figure 1. Sajag dashboard during the scripted digital-arrest scenario (CPU fallback on an x86 laptop; the same code uses the NPU on Snapdragon).

4. Why it must run on the Snapdragon NPU

- Privacy is the product.** A scam detector that uploads calls and screens would itself be a data-leak risk. Sajag keeps audio and frames in RAM and discards them.
- Always-on needs low power.** Screen OCR every 2 s and VAD every 32 ms all day is only viable on a dedicated NPU. Each workload is mapped to an HTP power mode (low_power_saver for sentinels, burst only while a call is live).
- Latency decides outcomes.** AI Hub publishes Whisper-Small at 117 ms (encoder) and 10.5 ms per token (decoder) on Snapdragon X Elite, with 100% of layers on the NPU, and Silero-VAD at 0.068 ms per 32 ms chunk. A warning lands while the caller is still speaking.
- Works offline.** No internet is needed after the one-time model download.

5. Technical implementation

5.1 Model stack and published NPU performance

Role	Model	Source	Snapdragon X Elite (AI Hub)	Snapdragon X2 Elite (AI Hub)
Voice gate	Silero-VAD (w8a16 mixed)	Qualcomm AI Hub	0.068 ms, 51/51 layers NPU	0.058 ms
Speech to text	Whisper-Small, multilingual	Qualcomm AI Hub	enc 117.1 ms, dec 10.5 ms/token, 100% NPU	enc 52.9 ms, dec 6.3 ms/token
Speech to text (X2 option)	Whisper-Large-v3-Turbo	Qualcomm AI Hub	enc 560.8 ms, dec 8.5 ms/token	enc 251.3 ms, dec 4.9 ms/token
Screen text	PP-OCRv4 det + rec	PaddleOCR via RapidOCR	static-shape fp32 graph, fp16 on HTP	same
Scam intent	Sajag classifier	trained in repo	[1,16384] Gemm + Softmax, 0.5	same

			MB	
Explanation (optional)	Llama-3.2-3B-Instruct w4a16	Qualcomm AI Hub (Genie)	11.3 tok/s, TTFT 0.12 s	42.5 tok/s

5.2 NPU engineering

- **Session factory (sajag/npu.py).** Every model is created through one factory that selects the QNN Execution Provider with the HTP backend (QnnHtp.dll), graph finalisation level 3 and fp16 precision, and falls back to the CPU provider elsewhere.
- **Static-shape adapter (sajag/vision/ocr.py).** The QNN EP rejects dynamic shapes. The OCR detector is pinned to [1,3,736,1280] and the recogniser to [1,3,48,1280]; the adapter resizes screens into the compiled canvas, zero-pads inputs, crops outputs and sends only oversize lines to the CPU. Verified to reproduce the dynamic-shape text on the demo screens.
- **Power-aware scheduling.** Workloads map to HTP performance modes: always_on to low_power_saver, background to power_saver, interactive to high_performance, burst to burst.
- **EP context caching.** The first launch compiles each graph for the HTP and stores an embedded context binary; later launches load it directly.
- **Strict NPU mode.** SAJAG_NPU_STRICT=1 sets session.disable_cpu_ep_fallback so unsupported operators fail loudly during development.
- **Torch-free Whisper.** AI Hub's reference decode loop (hf_whisper app) is ported to NumPy with a 200-slot static self-attention KV cache, a NumPy log-mel front end matching openai/whisper and a tiktoken tokenizer, so the runtime needs no PyTorch.
- **Compute gating.** A dHash screen gate, a call gate and the VAD keep the NPU idle most of the time.
- **Optional QDQ quantisation.** scripts/prepare_npu_models.py can emit w8a16 QDQ models with get_qnn_qdq_config; fp16 is the default because the synthetic calibration set was not representative enough.

5.3 Risk engine and fusion

Per sentence: text is normalised (NFKC, case, punctuation, common ASR/OCR confusions) and matched against 300+ cues grouped into 10 tactics (authority, threat, urgency, secrecy, payment, credential, remote_access, lure, fake_error, identity_bait). Latin phrases match on word boundaries, Devanagari as substrings, and the longest matching phrase becomes the evidence. The pattern score is a noisy-OR of tactic weights; the segment score combines it with the classifier probability as $1 - (1 - \text{pattern})(1 - 0.6 p)$. Genuine awareness messages ("police never arrest anyone over a video call", bank SMS footers) are damped.

Across the conversation: every tactic hit becomes timestamped evidence, weighted by the classifier's confidence in that sentence and decaying with a 4-minute half-life. The conversation score is a noisy-OR over the strongest evidence per tactic. Escalation rules encode investigator-described signatures: remote access plus OTP or money (floor 0.85), secrecy plus authority plus money (0.90, digital arrest), a prize or refund that must be paid for (0.68), plus a bonus when two modalities agree during a live call. Alerts fire once per upward level change.

6. Results

Evaluation	Result
Scripted end-to-end scenarios (real OCR on rendered screens)	5 / 5 reach the expected level
Digital-arrest scenario	WARNING at 20 s, DANGER at 26 s; money request at 45 s
Tech-support scenario	WARNING at 27 s, DANGER when AnyDesk starts at 35 s
200 synthetic scam calls	91% reach WARNING or DANGER; median 2 turns
200 synthetic benign calls with hard negatives	0.5% false alarms
Per family detection	digital arrest 100%, investment job 82%, kyc bank 100%, lottery prize 100%, refund upi 100%, tech support 84%, telecom utility 68%
Classifier alone, template-held-out	precision 1.00, recall 0.40 (why patterns and fusion are needed)
Tests	27 / 27 pass (python -m pytest)

Latency measured on an x86 CPU (fallback path)	classifier 0.03 ms, full text analysis 1.8 ms, OCR det + rec ~730 ms (NPU on Snapdragon)
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Honest limitations: the corpus is synthetic and its templates overlap the training data, so these figures are an upper bound; validation on consented real-call recordings is the first roadmap item. NPU latencies are quoted from Qualcomm AI Hub profiles; python -m sajak bench measures them on the device.

7. Deployment and accessibility

- **One-command install.** install.ps1 creates a Python 3.11 ARM64 environment, installs onnxruntime-qnn, downloads the AI Hub assets with qai_hub_models_cli, prepares the NPU OCR graphs and runs a doctor check. Judges without a Snapdragon PC can run python -m sajak demo or python -m sajak ui on any laptop.
- **Accessible by design.** English, Hindi and Hinglish in speech, screen and alerts; full-screen, large-type, spoken warnings designed for elderly users; no account or sign-in; works offline; open source under Apache-2.0.
- **Distribution.** OEM preload on Snapdragon HP AI PCs (HP leads India's PC market), bank and NBFC fraud-prevention programmes, I4C / 1930 awareness partners, senior-citizen associations, colleges and CSCs.
- **Impact.** At 2024 averages every 1,000 interrupted digital-arrest attempts protects about ₹15.5 crore, before counting tech-support, KYC and investment fraud.

8. How the idea was chosen

Fifteen candidate solutions were generated by mapping Qualcomm AI Hub's model catalogue to Indian problems where on-device processing matters, and by reviewing public repositories of other challenge entries to avoid crowded themes. Each was scored 1-5 on six criteria derived from the four official judging criteria, with weights from an AHP pairwise comparison (consistency ratio 0.013).

Rank	Candidate	Weighted score
1	Sajak: on-device multimodal scam shield (screen + call + system)	4.76
2	Deepfake video-call detector (face + voice liveness)	4.06
3	Offline crop-disease + Hindi voice advisor for CSC / FPO kiosks	3.74
4	Drone-video triage ground station for disaster response	3.67
5	Rural clinic medical scribe (ASR + note LLM)	3.57
6	Indian Sign Language to speech translator	3.57

Criteria weights: Technical depth on NPU 0.311, Innovation / novelty 0.192, India impact 0.171, Deployment & accessibility 0.136, Why on-device 0.090, Buildability & demo 0.101.

9. Innovation compared with existing tools

Microsoft Edge's scareware blocker and Google's on-device scam detection prove the direction (local AI against scams) but each covers one surface: browser pages or phone calls and messages. Caller-ID apps judge the number, not the conversation, and helplines act after the loss. Sajak is, to our knowledge, the first open tool that fuses call audio, screen content and system state on a Windows PC, in Indian languages, entirely on the NPU, and explains every alert.

10. Roadmap

- **Validation:** consented real-call pilot with bank or I4C partners.
- **Languages:** Marathi, Tamil, Telugu and Bengali cue libraries (Whisper already supports them).
- **Voice-clone detection:** synthetic-speech detector on the NPU using AI Hub's WavLM backbone.
- **Family guardian:** optional, consented alert to a trusted contact.
- **Phones:** the same AI Hub models on Snapdragon 8 Elite Android devices.

11. Originality and compliance

Sajag AI was designed and built by the participant during the challenge submission period; it is a new project, not a modification of prior work. It uses AI models from Qualcomm AI Hub and open-source platforms under their own licenses (Silero-VAD MIT, Whisper MIT, PaddleOCR / RapidOCR Apache-2.0, Llama 3.2 Community License). The code is released under Apache-2.0. The submission contains no confidential information or trade secrets. Demo screens are synthetic and watermarked as simulated.

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